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## **Pore-Scale Investigation of Halite Precipitation and Hydrate Formation During CO<sub>2</sub> Injection**

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### **Abstract**

Carbon dioxide (CO<sub>2</sub>) storage in deep saline aquifers and/or depleted hydrocarbon reservoirs is a widely recognized approach for reducing greenhouse gas emissions. However, two key phenomena, halite precipitation and CO<sub>2</sub> hydrate formation, pose significant challenges to maintaining injectivity and permeability near the wellbore. Halite precipitation occurs due to water evaporation (dry-out) into the injected CO<sub>2</sub> phase, leading to pore blockage. Hydrate formation, influenced by the low temperature of injected CO<sub>2</sub> and additional Joule-Thomson cooling due to CO<sub>2</sub> expansion, alters flow behaviour and reduces permeability.

This study provides novel experimental insights into how pore-scale influences porosity loss during CO<sub>2</sub> sequestration. By integrating microfluidic visualization with quantitative image analysis, the result improves the understanding of the coupled interactions among salt precipitation, hydrate formation, and porous media. The findings serve as a foundation for improving predictive models and refining injection strategies to mitigate permeability impairment, ultimately enhancing the efficiency of CO<sub>2</sub> storage operations.

A series of controlled microfluidic experiments using glass-based porous networks were conducted to observe the interactions between brine, CO<sub>2</sub>, and porous media under reservoir-relevant conditions. High-resolution imaging techniques, coupled with advanced image processing algorithms, were employed to analyse water film behaviour and salt crystal growth dynamics. The separated experiments explored the effects of varying pore structures, pressure fluctuations, and thermal conditions on the spatial distribution and morphology of hydrates. The impact of local water saturation variations on fluid displacement and hydrate stability was also examined.

The results indicate that heterogeneous pore networks retain more brine than homogeneous ones, leading to more salt precipitation and a maximum observed porosity reduction of 10%. Salt crystallization follows two distinct patterns: smooth-edged crystals form within the brine phase, whereas rough-edged deposits

develop at the CO<sub>2</sub>-brine interface. Hydrate formation exhibits diverse morphologies, amongst others pore-filling, grain-coating, and patchy, hydrate films, influenced by pore size, wettability, and pressure variations. The formed hydrates can reduce porosity by maximum of 15% in the experiments. Higher pressures and lower temperatures promote rapid hydrate nucleation, particularly in confined pores, while transient pressure pulses accelerate hydrate growth rates. Additionally, the spatial distribution of hydrates is found to be non-uniform, governed by fluid-phase interactions, with a weak correlation between hydrate and local water saturations.

**Keyword:** Microfluidic, Salt precipitation, CO<sub>2</sub> hydrate, Porosity reduction, CO<sub>2</sub> storage

## Introduction

Carbon capture and storage (CCS) in deep saline aquifers and depleted gas reservoirs has emerged as a critical approach for mitigating greenhouse gas emissions. However, the injection of CO<sub>2</sub> into these geological formations introduces complex physical-chemical interactions, such as halite precipitation and CO<sub>2</sub> hydrate formation. Both phenomena affect injection efficiency near wellbore regions (Darkwah-Owusu et al., 2024; Muller et al. (2009)). When dry CO<sub>2</sub> displaces formation brine, water evaporation occurs and results in the localized salt (halite) precipitation. The deposited halite crystals can fully or partially clog pore throats and substantially reduce reservoir permeability, which thereby impairs CO<sub>2</sub> injectivity and storage integrity (Ott et al., 2015; Pruess and Müller, 2009). Field-scale studies, such as the Quest CCS project in Canada, have documented severe injection impairments caused by near-wellbore salt accumulation and the necessity for effective mitigation strategies (Smith et al., 2022).

Simultaneously, CO<sub>2</sub> hydrate formation can occur when injected CO<sub>2</sub> meet connate water under specific conditions, specifically low temperature induced by Joule-Thomson cooling during injection in depleted gas reservoirs (Aghajanloo et al., 2024b). CO<sub>2</sub> hydrates are the crystalline solid compounds formed by CO<sub>2</sub> molecules trapped within water lattices and can rapidly nucleate and grow (Aminnaji et al., 2024). This process may reduce rock permeability and CO<sub>2</sub> injectivity, which has the potential to present further challenges to effective geological storage (Machado et al., 2023). Therefore, a deeper understanding of these pore-scale processes is crucial for optimizing injection strategies and improving long-term operational safety and storage efficiency.

In recent decades, microfluidic "lab-on-chip" technologies have been applied to separately study salt precipitation and CO<sub>2</sub> hydrate formation under reservoir-relevant conditions (He et al., 2022; Ho and Tsai, 2020; Ratanpara et al., 2025). Kim et al. (2013) tracked gas, brine, and salt phases in a geologically representative micromodel and observed a 20% porosity reduction due to salt deposition. At the pore-scale, two distinct salt crystallization modes were identified: pore-filling salt crystals forming early within trapped brine and finer polycrystalline salt aggregated forming later at receding brine-CO<sub>2</sub> interfaces. Nooraiepour et al. (2018) conducted a microfluidic study to examine how CO<sub>2</sub> phase states and injection flow rates influence salt precipitation in shale caprock fractures under reservoir conditions. They found that gaseous CO<sub>2</sub> at 20 cm<sup>3</sup>/min flow rate promoted significantly higher salt coverage up to approximately 11%, compared to liquid (~3%) and supercritical CO<sub>2</sub> (~0.8%), due to greater water evaporation with the gaseous phase. Additionally, lower injection flow rates (5 cm<sup>3</sup>/min) resulted in increased salt accumulation to 19% and 14% coverage for gaseous and scCO<sub>2</sub>, respectively, which was attributed to the higher chance for capillary-driven brine backflow and salt capillary suction. Yan et al. (2025) conducted comprehensive microfluidic experiments to investigate the pore-scale dynamics of salt precipitation during CO<sub>2</sub> injection into saline aquifers and illustrated the influence of pore structure heterogeneity and surface wettability. They observed two distinct salt crystallization types: smooth-edged crystals forming inside brine clusters and rough polycrystalline aggregation at the CO<sub>2</sub>-brine interface. Experiments revealed that heterogeneous pore structures retained more residual brine (initial water saturation ~45%) and caused nearly double the final salt

saturation (9.5%) compared to homogeneous structures (~5.3%). The study further documented capillary-driven water backflow near injection points, which sustained salt precipitation locally and highlighted the critical interplay between pore-scale flow, wettability, and heterogeneity in determining salt deposition patterns.

More recently, microfluidic technologies have been applied to study CO<sub>2</sub> hydrate formation under reservoir-relevant conditions. Some studies have focused on where and how CO<sub>2</sub> hydrate nucleation begins in porous media, which is critical for predicting hydrate formation (Dehghani et al., 2025; Morais et al., 2020; Wells et al., 2021). Observations confirm that hydrate nucleation is highly stochastic, but flow conditions can impose some spatial pattern. For example, Yu et al. (2025) found that actively injecting CO<sub>2</sub> hydrate crystals tended to nucleate near the CO<sub>2</sub> injection point, where local subcooling was strongest rather than randomly throughout the chip. Ouyang et al. (2025) used different micromodel geometries and identified two distinct nucleation modes: in low-permeability pore networks, they observed homogeneous nucleation with many random microscopic sites activating at once (a "stochastic multi-site" pattern), whereas in high-permeability networks, nucleation was heterogeneous occurring at the CO<sub>2</sub>-water interface, e.g. on the moving gas-liquid front. Microfluidics has also enabled quantitative measurements of hydrate growth rates and morphological evolution. One aspect is hydrate film growth at gas-water interfaces. Wadsworth et al. (2024) used a microfluidic chip with Raman imaging to study CO<sub>2</sub> hydrate films forming on channel walls (simulating a pipeline or wellbore scenario). They consistently observed a bi-layer hydrate structure: a dense "annealed" hydrate layer adjacent to liquid water and a more porous granular hydrate layer facing the CO<sub>2</sub> gas phase. By systematically varying conditions (subcooling 0.5~1.5 K, pressure 2.5~3.1 MPa, and CO<sub>2</sub> flow rate spanning 0.167 to 1.67 mm<sup>3</sup>/s), they found that higher CO<sub>2</sub> flow led to significantly thicker hydrate layers over the same time period, which suggests that convective gas supply (CO<sub>2</sub> influx) can control the hydrate growth kinetics in a diffusion regime.

While these studies have significantly advanced the understanding of individual phenomena, there remains a critical need for integrated research that examines both salt precipitation and hydrate formation within dynamic multiphase systems. Moreover, challenges such as limited dimensionality, simplified surface chemistry, and insufficient pressure-temperature control in microfluidic devices must be addressed to better replicate the complexity of reservoir environments (Li et al., 2020; Mim et al., 2023; Ratanpara et al., 2025). One key research question is how these two phenomena interact within porous media and, importantly, which process initiates first. When hydrates form in pores, the local salt concentration of connate brine increases due to migration of water molecules inside hydrate crystal structures. Once the concentration reaches the salt supersaturation, salt precipitation appears in the pores (Aghajanloo et al., 2024b). Under this regime, the probability of hydrate nucleation plays a significant role in this process. Hydrate nucleation events are random and exhibit unstable in the early stage of growth in microfluidics (Wang et al., 2021). In contrast, if salt precipitation occurs first, the high salinity generally inhibits the hydrate formation (Koh et al., 2002). However, precipitated salt crystals can provide favourable sites for hydrate nucleation (Khan et al., 2019). Therefore, which process dominates the early stage of CO<sub>2</sub> injection is crucial for mitigating the potential impairment on injectivity in near wellbore regions. Besides, after both salt and hydrates appear, it remains necessary to compare how and to what extent they partially or fully block pores in the porous medium. Additionally, the pressure fluctuation near wellbore, caused by operations such as shut-in or flow rate changes (Zhang et al., 2020), often induce unstable thermodynamic conditions for hydrates (Freij-Ayoub et al., 2007). Thus, studying the behaviour of salt and hydrates within the same pore structure is essential.

This study aims to investigate the two processes of halite precipitation and CO<sub>2</sub> hydrate formation during CO<sub>2</sub> injection at the pore scale, using advanced microfluidic techniques combined with high-resolution imaging. The experimental setup replicates conditions representative of low-pressure reservoirs, including precise control of pressure and temperature, to simulate the complex multiphase flow in porous media. By

observing salt and hydrate formation within the same pore network, the research seeks to clarify their own behaviour and determine their effects on the pore blockage during injection. Additionally, it evaluates the effects of pressure fluctuations and flow variations on the growth and distribution of these solid phases. The expected outcomes include a detailed understanding of how salt and hydrates impair permeability.

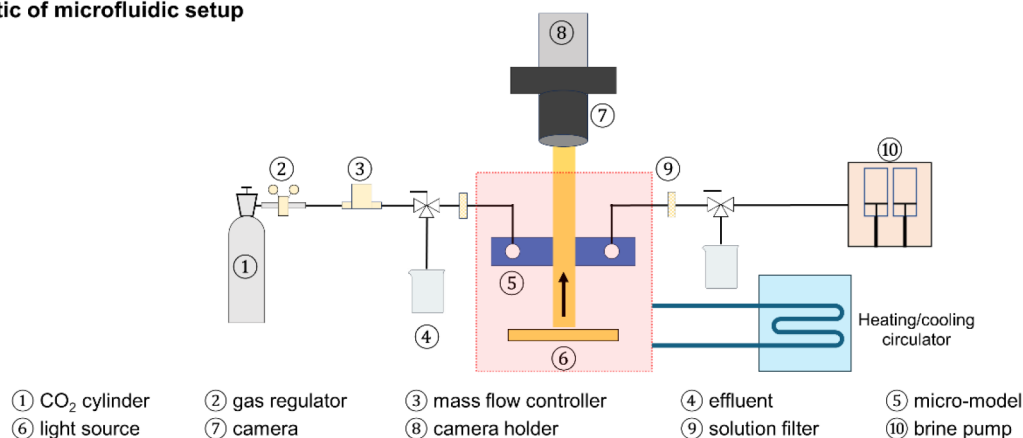
## Experimental approach

### Experimental setup and microfluidic property

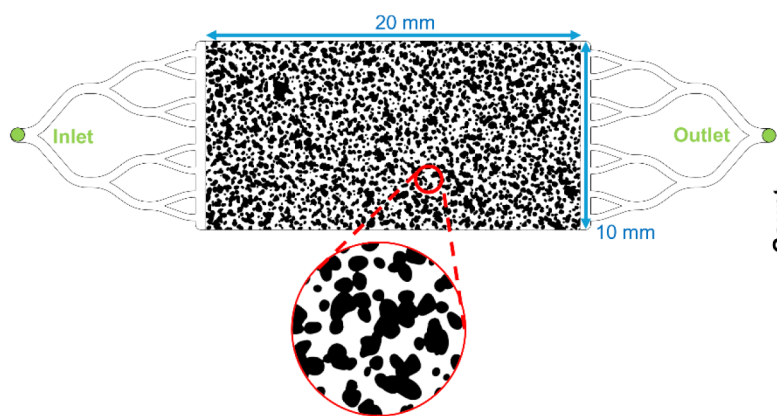
A custom-designed experimental setup has been assembled for this study in the laboratory as depicted in Fig. 1. The setup includes three main parts: CO<sub>2</sub> cylinder and brine supplement part from a pump, imaging part composited with camera and micromodel stage, and heating/cooling part controlled by a circulator and a sealed box. CO<sub>2</sub> gas is supplied from a high-pressure cylinder and regulated via a mass flow controller and multiple pressure regulators and valves. The injection flow rates can be control in the range of 0.007 to 0.7 ml/min. Fine filtration (0.25 µm) at both inlet and outlet lines prevents contamination. Brine or water injection is managed using a Quizix Precision Pumps. The microfluidic chip is mounted inside a high-pressure holder rated to 80 bar and enclosed within an isolation box with three temperature sensors continuously monitoring temperatures at room, chamber and inside chip. The temperature is maintained through a cooling bath and a thermally insulated chamber within  $\pm 0.2$  °C variation. Two pressure sensors are inserted in the inlet and outlet lines with a maximum pressure of 80 bar and accuracy of 0.01 bar. The imaging system comprises a Canon EOS 90D camera body paired with an EF 100mm f/2.8L Macro IS USM lens, which offers an observation domain of 22.3 mm (length)  $\times$  14.9 mm (width) with a resolution of 3.7 µm/pixel. The maximum magnification of the lens is 5:1. This combination allows us to observe the whole porous area of the chip.

The microfluidic chip is fabricated by Micronit Micro Technologies and constructed using borosilicate glass and fused silica materials. The porous domain is achieved through an acid chemical etching method and contains isotropic etching features. The domain has a dimension of 20 mm (length)  $\times$  10 mm (width)  $\times$  0.02 mm (depth), as shown in Fig. 1 (b). The pore structure is designed for mimicking the grain structure of physical rocks. Fig. 1 (c) gives the pore size distribution. The chip has a range of pore size from 75 µm to 675 µm with a geometric median value of 280 µm. The porosity and permeability are 0.56 and 6.9 Darcy, respectively. As the chip is made by the silica glasses, the wettability of the pore is water-wet with a measured contact angle 40° of water-air.

## (a) Schematic of microfluidic setup



## (b) Structure of two micro-chips



## (c) Pore size distribution

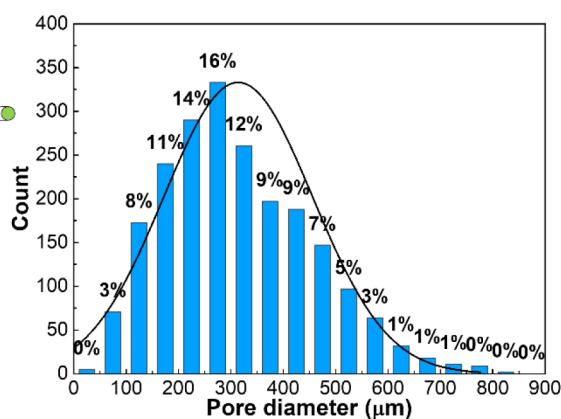


Figure 1—Schematic of experimental setup and characterization of the microfluidic porous medium. (a) The schematic diagram of the microfluidic experimental setup includes the major components marked with numbers; (b) The detailed design of the microfluidic chip shows the porous medium region with a zoomed-in view revealing the complex pore structure. The porous domain is 20 mm in length and 10 mm in width and contains a physical rock structure; (c) The pore size distribution histogram of the porous region displays the variability of pore sizes fitted with a normal distribution curve.

## Experimental procedure and conditions

**Salt precipitation experiment.** Due to the limitations of our current setup, it is still challenging to achieve the simultaneous observation on salt precipitation and hydrates in one experiment. Two main challenges are 1) differentiating between solid salt and hydrates, and 2) well-controlled low temperature (e.g.  $-25^{\circ}\text{C}$ ) for hydrate formation with a high salinity brine (25 wt% salt concentration). Therefore, we separately investigate these two processes and evaluate their own effects on the porous domain. For salt precipitation experiment, 25 wt% NaCl solution is chosen to prevent the extra chemical reactions between CO<sub>2</sub> and chemicals in brine solutions. The experimental temperature is controlled around  $50^{\circ}\text{C}$  and pressure is the atmospheric pressure. The brief procedure and conditions are illustrated as follows:

- Preparation stage: The microfluidic chip is thoroughly cleaned to remove surface dust and contaminants that might affect imaging quality. Camera and lens settings are carefully optimized to achieve sharp focus on the pore regions for precise visualization.
- CO<sub>2</sub> saturation stage: The inlet and outlet tubing are connected to the chip, and CO<sub>2</sub> is injected at a controlled flow rate of 0.01 ml/min for a duration of 10 minutes to reach full saturation of the porous network.

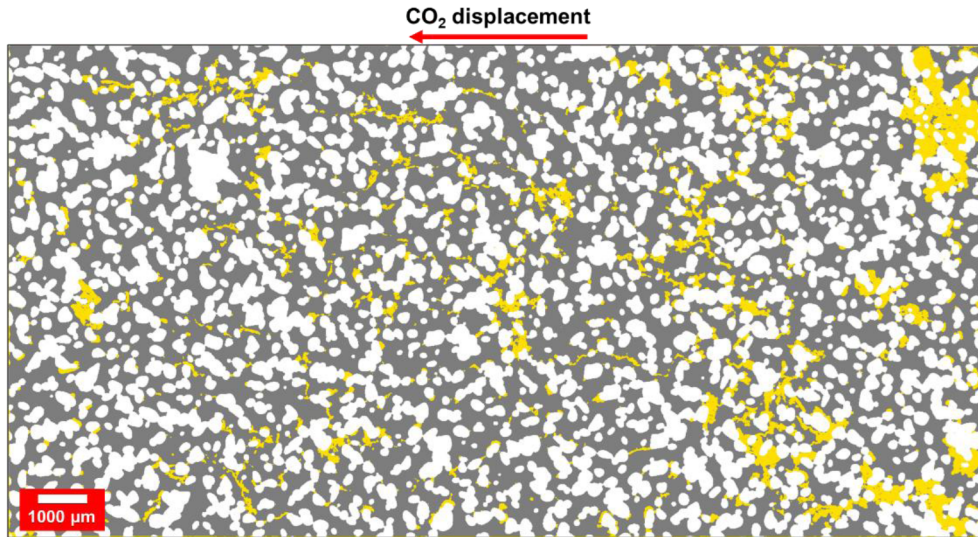
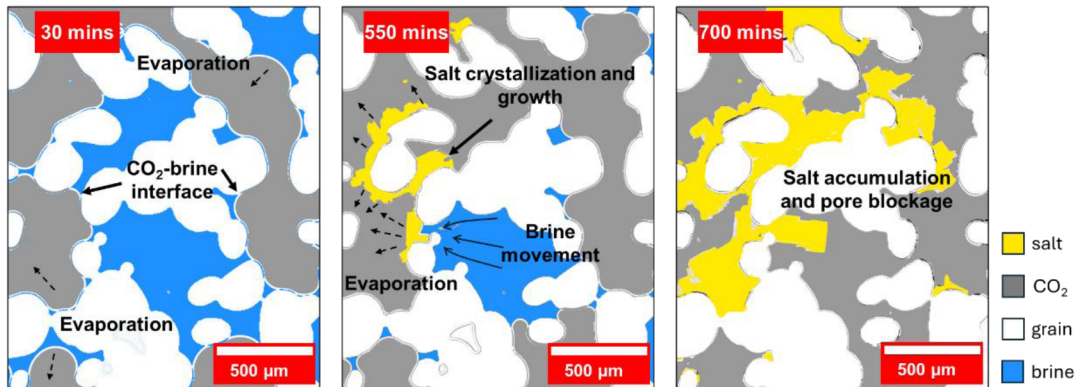
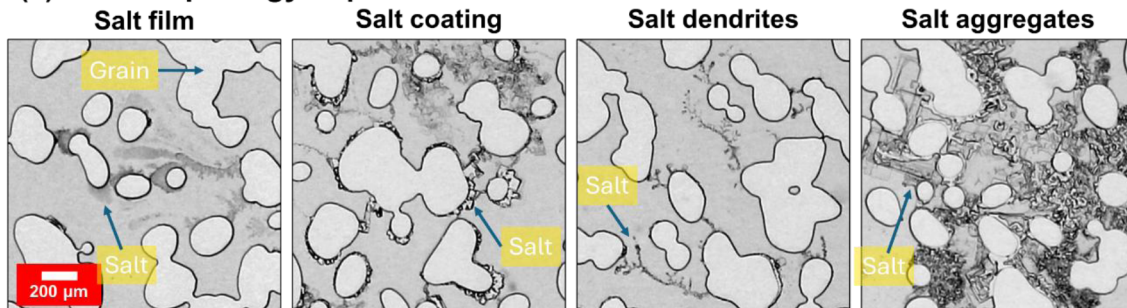
- Brine saturation stage: The valve to the CO<sub>2</sub> mass flow controller is closed and redirected to the outlet line, then start the continuous injection of brine at 0.01 ml/min until no gas bubbles are observed within the porous medium, which means the completed brine saturation.
- CO<sub>2</sub> injection stage: The CO<sub>2</sub> injection line is then reopened and connected to the outlet effluent, and CO<sub>2</sub> injection resumes at a controlled flow rate of 0.01 ml/min regulated by the mass flow controller throughout the whole experiment. After finishing CO<sub>2</sub> breakthrough, the imaging properties, e.g. frame speed and imaging time, are set and kept the monitoring until the end of the experiment. The heating bath is set the temperature to 50 °C for accelerating the water evaporation.
- Cleaning stage: Upon completion of the experiment, the chip is cleaned by sequentially flushing with deionized water. The chip is subsequently dried in a vacuum oven at 70 °C for 30 to 60 minutes to prepare it for further use.

**Hydrate formation experiment.** The procedures for hydrate experiments have similar processes as those in the salt precipitation experiment, including the preparation stage, CO<sub>2</sub> saturation stage, brine saturation stage and cleaning stage. The difference is that, to achieve the hydrate formation, the temperature and pressure must reach the favourable conditions for hydrate formation. And those conditions need to be controllable for avoiding the notable variations to influent hydrate stability. In our experiment, with using demi-water, the temperatures and pressures for hydrate formation and dissociation are 0.5 °C (sub-cooling 6.5 °C) and 30 bar for hydrate formation stage and 15 °C and 30 bar for hydrate dissociation stage. Due to the confined space of microfluidic and the limited contact interface of CO<sub>2</sub>-water, a pressure perturbation is commonly introduced into the microfluidic system to trigger hydrate formation or shorten the induction time (Fukuyama et al., 2024; Ouyang et al., 2023; Zhang et al., 2024b). In our experiments, we control the variation of pressure perturbation within 3 bar through an electrical valve connected in the inlet line. During the stage of CO<sub>2</sub> saturation, the pressure is maintained at 30 bar in the whole system. After CO<sub>2</sub> breakthrough, the temperature inside the isolation box is decreased to 0.5 °C. When the temperature and pressure are stable, we normally wait for at least 1 day for full CO<sub>2</sub> dissolution into connate water. If there is no sign of hydrate in the monitoring images, the pressure perturbation can be applied to the system. Although the pressure variation induces the CO<sub>2</sub> and water movement, the induction time is shortened, and it would not affect the hydrate formation and growth. For hydrate dissociation, we use the thermal method to slowly increase the temperature by 15 °C to over the hydrate equilibrium conditions.

## Results and Discussion

### Morphology of precipitated salt

To quantitatively analyse the dynamic process of salt precipitation, the images and data are processed using ImageJ and MATLAB. ImageJ is employed for initial image segmentation and thresholding, which can precisely distinguish CO<sub>2</sub>, water, salt crystals, and grains. Subsequently, MATLAB scripts analyse these segmented images to compute key morphological parameters, such as crystal size distribution, and porosity change. Fig. 2 provides a detailed illustration of salt precipitation dynamics within a porous medium under controlled flow conditions.

**(a) Segmented image at t=16 hours****(b) Dynamic salt precipitation in pores****(c) Salt morphology in pores**

**Figure 2—Salt distribution map and morphology evolution in a microfluidic porous medium under dynamic flow conditions.** (a) Segmented image after salt precipitation in the whole porous domain of microfluidic chip, which shows the spatial distribution map. Colours present four phases: salt (yellow), solid grains (white), gas phase (grey), and brine (blue). The  $\text{CO}_2$  flow direction from right to left is indicated by the red arrow on the top. (b) Three segmented images at 30, 550, and 700 minutes after  $\text{CO}_2$  breakthrough to illustrate the progressive salt crystallization and pore space changes, which illustrates the transition from brine-filled pores to extensive salt blockage. (c) Zoom-in 8-bit raw images displaying diverse salt crystal morphologies observed within the porous network, including continuous films, grain coatings, dendritic structures, and patchy aggregates. Scale bars indicate spatial dimensions.

Fig. 2 (a) shows the final phase distribution, including glass grains (grey solids), gas  $\text{CO}_2$  (white regions) and salt deposits (yellow). Notably, large yellow clusters appear near the right (inlet) side, which means that the salt precipitated close to the inlet region. This spatial distribution aligns well with findings from both core flooding and microfluidic experiments reported in the literature (Ott et al., 2015; Sokama-

Neuyam et al., 2023; Zhang et al., 2024a), where salt accumulation commonly occurs near the injection inlet due to localized brine evaporation and capillary-driven brine backflow (Miri and Hellevang, 2016). The pore structure and connectivity significantly influence the spatial distribution and morphology of salt precipitation. Although salt accumulation is more pronounced near the inlet, as shown in the figure, certain pores in other regions develop large clusters. This phenomenon has been captured and explained in our previous paper (Yan et al., 2025). Additionally, the salt deposits in the central and left regions of the chip predominantly exhibit band-like formations, which indicates that salt accumulation in these areas is more distributed rather than large clusters under the current experimental conditions. These salt precipitation patterns are governed by multiple factors, including surface wettability, pore heterogeneity, brine composition, and flow rate (Sun et al., 2024).

Fig. 2 (b) illustrates the dynamic process of salt precipitation within pores at three distinct time points: 30 minutes, 550 minutes, and 700 minutes. At the initial stage (30 minutes), the pores are predominantly saturated with connate brine, which exists as discrete brine clusters, trapped brine within dead-end pores, and brine collars surrounding the grain surfaces. The CO<sub>2</sub>-brine interfaces indicate that the grain surfaces exhibit strong water-wet characteristics. As water evaporation progresses, these interfaces gradually retract toward the brine phase. By 550 minutes, salt starts to nucleate at the solid grain surface. Notable salt crystallization and growth occur along the CO<sub>2</sub>-brine interfaces and grains. With ongoing evaporation from the interfaces and the wet salt crystals, salt capillary suction drives the brine movement to the formed salt which lead further crystallization and growth of salt. At 700 minutes, salt accumulation is extensive with significant pore blockage among the pores as shown as the large yellow region. From the salt positions, it can be seen that the large amount of brine tripped brine inside the centre pores move to the left edge of the pores instead of staying inside and aggregated to the large salt crystals.

Fig. 2 (c) presents four types of pore-scale salt crystal deposits observed in a microfluidic experiment: salt film, salt coating, salt dendrites, and salt aggregates. In some regions, salt forms thin, continuous coatings along channel walls. These film-like precipitates occur where brine left behind very thin wetting layers on surfaces. This morphology typically forms because of brine evaporation, where water migrates along pore walls due to capillary forces, leaving behind a crystalline salt layer once the dissolved salts reach supersaturation. The wettability of the pore surfaces plays a critical role in promoting this film formation with hydrophilic surfaces enhancing brine transport and subsequent salt deposition (Dąbrowski et al., 2025). Although rarely quantified in CO<sub>2</sub>-brine experiments, analogous "efflorescent films" are known to appear when evaporation leaves only trace liquid on grains (Derluyn and Prat, 2024). Salt coating is a morphology of pore-lining salt crystals that form a ring along the solid grains. The behaviour is similar as that salt crystals can remain pinned to a surface and spread as a porous salt coating along that substrate (Xia et al., 2025). In porous media, this refers to a partial but relatively uniform coverage of the pore surfaces by salt. Such a grain-coating precipitate might result from capillary films delivering solute to the grain boundaries and some crystals form few bridges between pores.

Dendritic salt crystals are observed in the microfluidic chip, appearing as tree-like branched structures extending into the pore space. These formations are characteristic of rapid evaporation and high supersaturation conditions (Ho and Tsai, 2020). Dendritic morphology starts as skeletal branches at the grain surface and rapidly grows outward from the brine-gas interface into the pore, due to the capillary forces of salt enhancing ion transport to the crystal growth fronts. Although such structures cannot significantly block the pore spaces in this work, those have the potential to bridge the pores that provides extra pathways for brine movement. Salt aggregates refer to irregular and discontinuous clusters of salt crystals that form unevenly along pore surfaces or within pore spaces. These patchy accumulations result from localized supersaturation zones where salt nucleation and growth occur preferentially through the porous structures driving brine movement through brine thin film (Wijnhorst et al., 2025). Unlike the salt films or large monocrystals, salt aggregates exhibit a heterogeneous spatial distribution, which creates scattered areas of

pore blockage that can significantly impact permeability by partially obstructing flow paths or completely sealing pores. Such morphologies have been frequently observed in microfluidic and core-scale studies of salt precipitation during CO<sub>2</sub> injection ((He et al., 2024; Li et al., 2024)), where they are linked to non-uniform drying fronts and variable brine connectivity.

Fig. 3 illustrates the distributions of water and salt distributions along x-axis of micro-chip and their saturations in the whole porous domain. Initially, pore area is relatively evenly distributed before CO<sub>2</sub> displacement, as shown in Fig. 3 (a). After CO<sub>2</sub> displacement, the connate brine area is smaller near the inlet (approximately  $2 \times 10^5 \mu\text{m}^2$ ) compared to the outlet region (around  $6 \times 10^5 \mu\text{m}^2$ ), due to the capillary-end effect at the outlet. As water evaporates, a drying front develops and gradually moves toward the outlet over time at 10, 12, and 14 hours. This drying front marks the transition zone where water evaporates and salt precipitates, which is consistent with the processes of evaporative concentration and salt crystal nucleation under supersaturation conditions. After the salt dry-out is complete, the area distribution of salt is shown in the bottom layer and presents a non-uniform distribution across the microchip. Notably, salt accumulates more near the inlet rather than in the initially higher brine saturation region near the outlet, which indicates that brine migrates through brine thin film and concentrates near the injection point due to the capillary-driven brine backflow. Fig. 3 (b) presents a quantitative comparison of dynamic water and salt saturations during the experiment, where shows three stages: slow evaporation, salt precipitation, and final stabilization. Water saturation (blue circles) starts at about 32% and decreases slightly due to slow evaporation into CO<sub>2</sub> at a rate of  $2.5 \times 10^3 \mu\text{m}^2/\text{min}$ . Around 10 hours, as the salt dry-out front emerges, water saturation drops sharply at a rate of  $7.1 \times 10^4 \mu\text{m}^2/\text{min}$ . This enhanced water loss results from both evaporation and conversion of water into salt crystals, where the water area suddenly diminishes while salt appears. Simultaneously, salt saturation (yellow circles) increases at a rate of  $2.4 \times 10^4 \mu\text{m}^2/\text{min}$ . In the final stage, both water and salt saturations gradually plateau and stabilize around 10%.

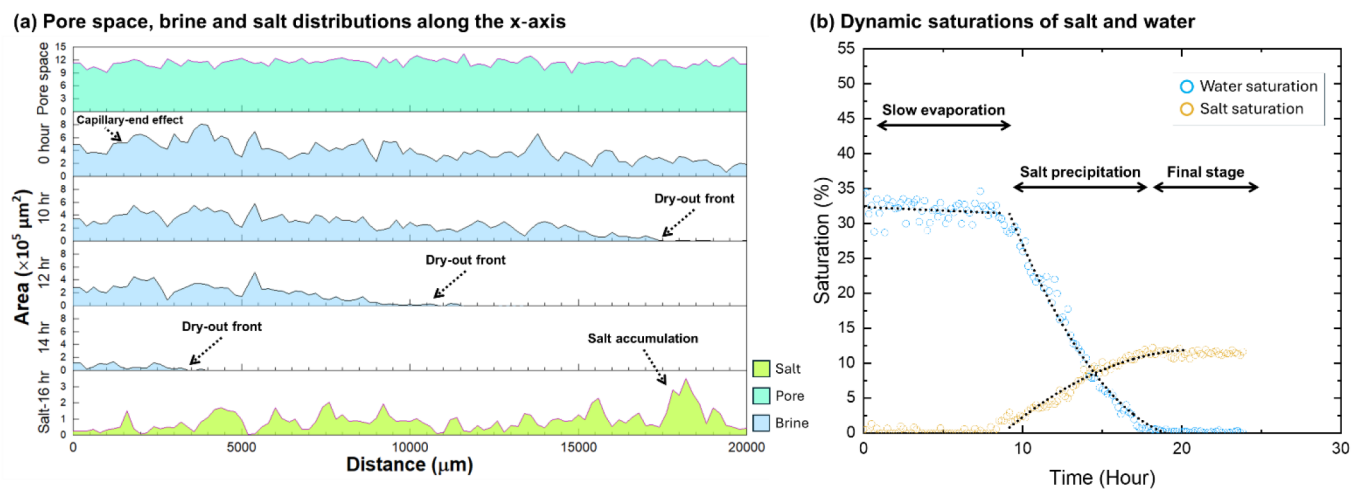


Figure 3—The dynamic process of salt precipitation in the whole porous domain. (a) the areas of pore, water and salt distributions across pore network. The top layer indicates the pore space area at before CO<sub>2</sub> displacement. The middle layers in blue colour illustrate water areas at different times after CO<sub>2</sub> displacement, 0, 10, 12, and 14 hours. The bottom layer in yellow presents the salt area at 16 hours. (b) The dynamic saturations of water and salt during the experiment marked with blue circles and orange circles.

## Morphology of hydrates

Though the experiment for hydrate, we can analyse hydrate morphologies and their impact on the pore structure of porous media. Using microfluidic techniques combined with high-resolution imaging, hydrates are observed to form predominantly along gas-liquid interfaces and around grain surfaces, which presents diverse morphologies such as clustered aggregates, thin films coating grains, and pore-bridging networks.

Fig.4 show the segmented image for phase distributions, including CO<sub>2</sub>, grain, water and hydrate, and the dynamic

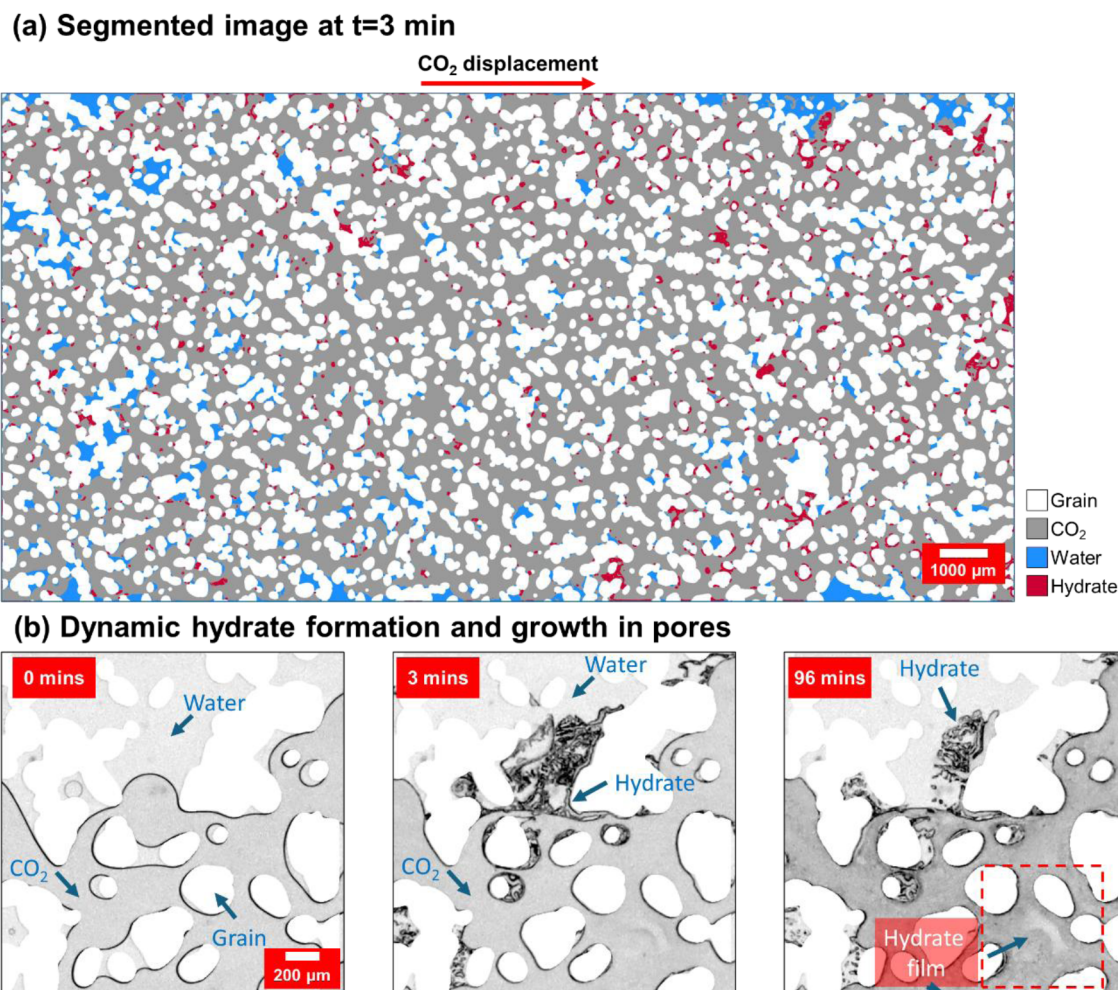
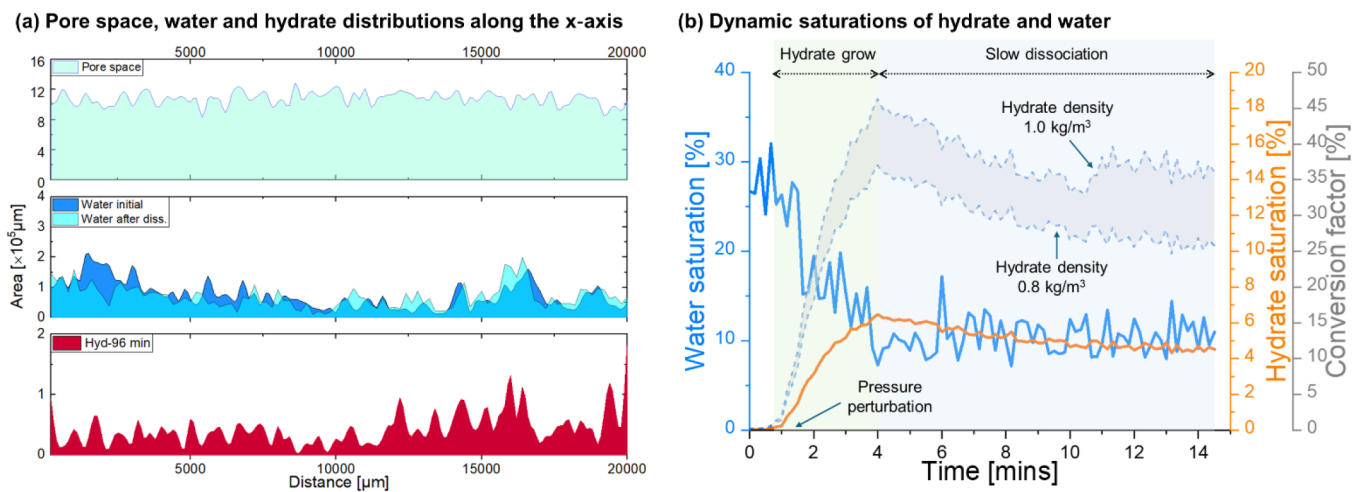


Figure 4—Phase distributions in the microchip and morphology evolution of hydrate formation in microfluidic porous media. (a) Phase segmentation map showing grain (white), water (blue), CO<sub>2</sub> gas (grey), and hydrate (red) distributions at pore scale. (b) Three 8-bit raw images captured at 0, 3, and 96 minutes depicting the hydrate growth.

Fig.4 (a) illustrates the phase distribution within the microfluidic porous medium. The connate water initially has saturation around 27% and mainly exists as collar water around pores and trapped water inside pores. After applying pressure perturbation, hydrates predominantly nucleate and grow along grain boundaries and within the pores. Some hydrates form clusters that bridge adjacent grains, which can fully or partially block pores. Other hydrates form and grow in the dead-end pores which contained connate water. In general, the formed hydrates distribute at location where connate water resides, which is unlike the salt accumulation close to the inlet region. This indicates that the hydrates do not induce the significant water movement under this water saturation. Fig.4 (b) presents optical micrographs showing the dynamic evolution of CO<sub>2</sub> hydrate morphology within a microfluidic porous medium. Initially, hydrate nucleation occurs as discrete clusters at the gas-liquid interface and along the isolated grains, where CO<sub>2</sub> dissolution and temperature gradients favour crystal formation. After hydrate formation for 3 mins, these isolated clusters grow inside water and coalesce into continuous. Some hydrates that have the interconnected water films coat the surfaces of grains. Such hydrate formation has been reported in prior research, including Maruyama et al. (2024), who demonstrated that film thickening is controlled by water molecule diffusion through the hydrate layer, and Martinez et al. (2021), who observed hydrate morphologies ranging from dendritic to

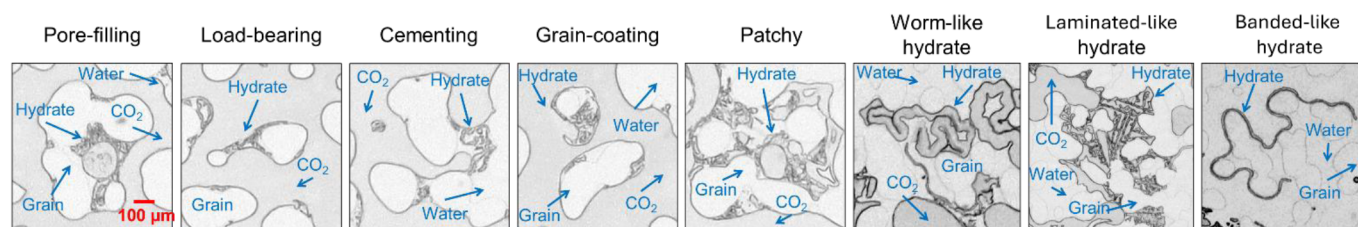
skeletal forms depending on thermodynamic conditions. At 96 minutes, parts of hydrate clusters dissolve, whereas larger hydrate accumulations maintain their morphology and spatial positioning within the pores. The phenomenon is because of the dynamic process of the  $\text{CO}_2$  consumption encapsulated in the hydrates (Li et al., 2021). After the  $\text{CO}_2$  is entirely consumed, hydrate crystals disappear and become transparent in the pore. Notably, hydrate films progressively develop around grain surfaces, which are visible as slightly darker parts as shown in the red-dashed region. This is consistent with literature that reports film-like hydrate structures forming due to interfacial energy minimization and grain surface wetting properties (Fukuyama et al., 2024).

Fig. 5 (a) illustrates the phase distributions along the x-axis of the microchip. The top layer shows a relatively uniform pore space distribution along the microchannel, as shown in the salt experiments. The middle layer compares the water areas before hydrate formation and after hydrate dissociation. After dissociation, water maintains a similar trend of distribution as initial state, which indicates that the connate water has no significant change. The water distribution also presents that there is no obvious water movement under this initial water saturation and conditions. The bottom layer depicts the hydrate distribution at 96 minutes, which shows the localized hydrate accumulation that corresponds to areas of water loss. These observations align with findings from Ji et al. (2021) where demonstrated that hydrate formation preferentially occurs in water-rich zones and impacts pore connectivity. Fig. 5 (b) quantitatively tracks the evolution of water and hydrate saturations and the hydrate conversion factor over time. The trend of hydrate saturation follows a two-stage trend: rapid hydrate growth followed by a slower dissociation. Initially, water saturation starts high ( $\sim 30\%$ ) and gradually decreases as hydrates nucleate and grow until reaching a steady state about 10% during the slow dissociation phase. As mentioned above, the dissociation is attributed to the gradual  $\text{CO}_2$  consumption inside hydrates. Correspondingly, hydrate saturation rises rapidly with a rate of  $2.3 \times 10^6 \mu\text{m}^2/\text{min}$  and reach the peak around 7%, then has a slow drop with a rate of  $-1.9 \times 10^5 \mu\text{m}^2/\text{min}$  until stabilizing at around 4.5%. Dynamic conversion factor can be evaluated as shown in the grey shadow. It rises rapidly at the early stage of hydrate formation, peaking maximum around 45% with two hydrate densities ( $0.8$  and  $1.0 \text{ kg/m}^3$ ) influencing growth rates. As the experiment progresses, the conversion factor begins to decline slightly, stabilizing between 25%.



**Figure 5—The dynamic process of hydrate formation in the whole porous domain. (a) Spatial distribution of pore space, water area before hydrate formation and after hydrate dissociation, and hydrate area at 96 minutes within a microfluidic porous medium (b) The dynamic changes of water saturation (blue curve), hydrate saturation (orange curve), and hydrate conversion factor (grey area).**

Fig. 6 presents the diverse morphologies of hydrates formed in porous media over time. We have observed the four phases in various contact in the pores, water, CO<sub>2</sub>, hydrate and grain. Pore-filling hydrate forms within the dead-end pore spaces. This structure may appear dispersed and lack structural connectivity with the pore network but resulting in high local hydrate saturation from the high local saturations of gas and water. These hydrates primarily form when gas molecules diffuse into pore spaces and subsequently nucleate. This morphology is particularly common in fine-grained porous media or where water is evenly distributed throughout the pore network (Lv et al., 2021). As pore-filling hydrates expand, hydrates obstruct pore spaces and lead to reductions in permeability. Load-bearing hydrates grow as part of the grain framework, creating a structure that supports sediment weight and resists compaction under high-pressure conditions. The nucleation is slower than pore-filling hydrate but steady growth under high pressure and low temperature (Ding et al., 2022). Growth kinetics in load-bearing hydrates are influenced by limited gas availability. Cementing hydrates are typically found at the contact points between grains, where it acts as a bonding agent that significantly enhances the mechanical strength of hydrate-bearing sediments (Malagar et al., 2019). The hydrates precipitate at the water-gas interface around grains and gradually cement adjacent particles together and form a stable structure. Grain-coating hydrates appear as a coating around individual grains and on gas bubbles, which are formed primarily under low driving forces and stable gas-water interfaces (Fukuyama et al., 2024). Limited nucleation sites mean that hydrate growth remains localized to grain surfaces. These commonly form in heterogeneous pore structures or under local lower saturation levels around grains.

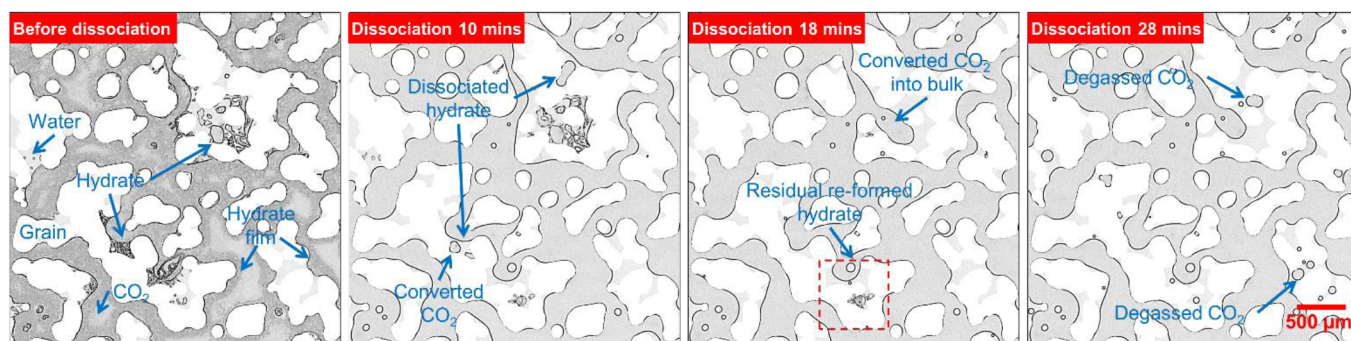


**Figure 6—8-bit raw images of eight hydrate morphologies. The phases are marked with arrows in the images. The captured hydrate morphologies are, pore-filling, load-bearing, cementing, grain-coating, patchy, worm-like, laminated-like, and banded-like.**

Patchy hydrates are characterized by their irregular and non-uniform distribution within the pore space. When gas and water contact is uneven across the porous matrix or fluctuations in pressure and temperature occur in the areas resulting in localized CO<sub>2</sub> supersaturation, these hydrates form in clusters or patches rather than in a continuous layer (Dai et al., 2012). These conditions create high driving forces and induces the high nucleation rates in specific regions, which gives the formation of isolated hydrate clusters. This nucleation can be intensified by Ostwald ripening, where smaller hydrate clusters dissolve and larger ones grow then lead to concentrated patches. The porous domain and grain surface have the potential to enhance hydrate dissolution and Ostwald ripening (Zhang et al., 2021). Worm-like (intestine-like) and laminated-like hydrates represent the elongated and branched hydrate structures, which often form along gas pathways or pore throats under conditions of rapid nucleation. During the nucleation phase, a dendritic pattern may initially form at the gas-water interface. As the driving force (e.g., pressure or temperature perturbation) increases and creates a temporary CO<sub>2</sub> supersaturation in water, nucleation is intensified and produces more complex structures such as the intestine-like morphology, then grows into the aqueous phase as nodules. The driving force from the perturbations overcomes any initial kinetic barriers to nucleation and causes rapid hydrate growth and unique morphologies (Lv et al., 2021). Additionally, the confined channels within micromodel provide gas favourable pathway where gas is plentiful for the preferable hydrate formation into these distinct growth patterns based on spatial constraints. When the temporary perturbations disappear, the driving force decreases and renders some hydrate structures unstable even dissociation. Interestingly, we

have observed that in the deeper water regions, where the contact with the gas phase is limited, hydrate formation rates are lower, and these hydrates are less stable and have dissociation. The banded-like hydrates (or hydrate shell) are first-time observed during the hydrate dissociation. We also notice that the worm-like, laminated-like and band-like hydrates are the first-time to be captured in one experiment at the same time. When the temperature increases,  $\text{CO}_2$  expansion occurs and induces water movements. These movements make new hydrate nucleation and growth around gas bubbles or along the grain surfaces, forming concentric shells. This morphology is linked to processes of water invasion, where water contacts with gas-rich zones in narrow pore spaces, nucleation begins at the interface and results in hydrate shells that encapsulate gas pockets or solid grains.

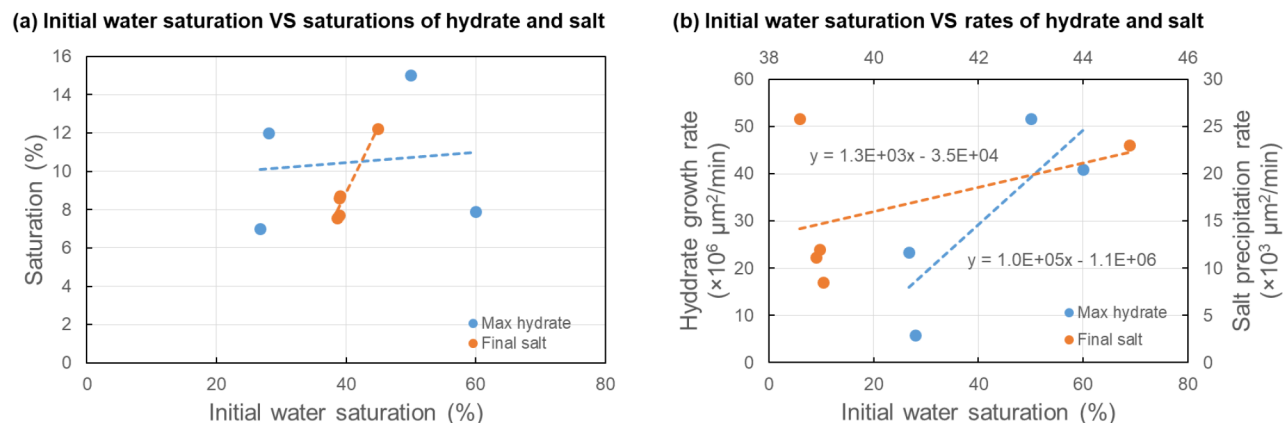
Fig. 7 captures the dynamic hydrate dissociation process at four sequential time points within a zoomed-in region of the microfluidic chip. Prior to dissociation, hydrates predominantly form as films or clusters attached to grain surfaces and occupy water-filled pores, particularly at the  $\text{CO}_2$ –water interfaces, which is consistent with earlier observations of hydrate morphology and spatial distribution in literature (Habiburrahman et al., 2025; Zhang et al., 2025). The film-like hydrate structures result from the interplay of capillary forces driving water thin film and interfacial energy minimization, which promotes lateral expansion along grain surfaces. We heat the system up to around  $7.5^\circ\text{C}$  that is near the thermodynamic conditions for the hydrate dissociation and maintain for 20 mins. After 10 mins of dissociation, hydrates adjacent to grain surfaces and interface regions start to dissociate first and release  $\text{CO}_2$  gas into the pore space. This released  $\text{CO}_2$  can nucleate bubbles within the aqueous phase or break the water film and merge into the bulk gas phase. At 18 mins, we find that hydrates residing in isolated dead-end pores dissociate more slowly or partially due to limited heat transfer and restricted mass exchange with the bulk fluids. This indicates that the heterogeneity of pore networks can induce the various capillary pressure and may inhibit the local hydrate dissociation (Sun and Mohanty, 2006). The local condition, including pressure, water and  $\text{CO}_2$  contents, can induce the re-formation of hydrate, as shown in the red-dashed square. This suggests the hydrate formation and dissociation are dynamic process and highly depends on the local thermodynamic conditions. With increasing temperature by  $9^\circ\text{C}$  at 28 mins,  $\text{CO}_2$  degassing occurs due to the decreasing dissolution with heating and tiny  $\text{CO}_2$  bubbles gather to obvious bubbles inside water.



**Figure 7—8-bit raw images of hydrate dissociation. morphologies. Hydrates (dark regions) initially form films and clusters on grain surfaces and within pore water (0 min). Progressive dissociation occurs over 10, 18, and 28 minutes, with hydrates near interfaces decomposing first and releasing  $\text{CO}_2$  gas bubbles.**

Fig. 8 presents the relationship with linear fits between initial water saturation and two key param: the saturation levels of salt and hydrate (left graph), and their respective growth rates (right graph). The left plot reveals that maximum hydrate saturation slightly increases with higher initial water saturation but not a significant trend. This indicates that greater water availability promotes hydrate formation within the porous medium, but the promotion is very limited. In contrast, the final salt saturation shows a strong positive correlation with initial water saturation, which suggests that salt deposition is more directly influenced by water content. The right plot compares hydrate growth rates and salt precipitation rates, which can be useful

for the comparison of the kinetics of salt and hydrate growth in porous media. The results show that hydrate growth rates exhibit a stronger dependence on initial water saturation compared to salt precipitation rates. Specifically, the hydrate growth rate increases rapidly as initial water saturation rises, which reflects the enhanced nucleation and crystal growth in water-rich conditions. Conversely, salt precipitation rates increase more moderately (100 times slower than the hydrate growth rate), implying that factors other than water saturation, such as brine salinity and flow dynamics, play significant roles in salt formation. Additionally, the hydrate growth can be more sensitive and significantly affected by the local thermodynamic conditions, while the salt precipitation rate is relatively stable that mainly depends on the water evaporation rate under specific conditions.



**Figure 8—Comparison of salt and hydrate in porous media. (a) Relationship between initial water saturation and the saturations of final salt precipitation and max hydrate formation. (b) Relationship between initial water saturation and kinetic rates for salt precipitation and hydrate growth. The blue circles present hydrate and orange circles present salt.**

## Conclusions

This study provides novel experimental insights into the pore-scale dynamics of halite precipitation and  $\text{CO}_2$  hydrate formation during  $\text{CO}_2$  injection under reservoir conditions. Using home-made microfluidic devices combined with high-resolution imaging and quantitative image analysis, we systematically characterized the morphology, spatial distribution, and dynamic saturations of salt and hydrate phases within complex pore networks. Our results reveal that the heterogeneous pore structures significantly influence brine retention, which causes salt precipitation with distinct morphologies such as film-like, coating, dendrites and aggregates. The smooth salt crystal and rough polycrystalline aggregates at  $\text{CO}_2$ -brine interfaces are captured. Some aggregated salt crystals have fully blocked the adjunct pores and prefer to accumulate at the regions close to the inlet. We observed a maximum porosity reduction of approximately 10% overall the whole microchip due to salt precipitation. Furthermore, the final salt saturation shows a strong positive correlation with initial water saturation and salt precipitation rates range from  $10^3$  to  $10^4 \mu\text{m}^2/\text{min}$ , which reflects evaporation-driven and capillary-induced salt migration mechanisms (Ott et al., 2015; Yan et al., 2025).

In contrast,  $\text{CO}_2$  hydrate formation mainly takes place at gas-liquid interfaces and grain surfaces, forming varied eight morphologies such as pore-filling, load-bearing, cementing, grain-coating, patchy, worm-like, laminated-like, and banded-like, which is strongly governed by local water saturation, wettability, and pressure perturbations. Maximum hydrate saturation exhibits a weaker correlation with initial water saturation (up to  $\sim 15\%$ ) and shows that hydrate formation is more strongly influenced by local thermodynamic conditions and  $\text{CO}_2$  availability. Hydrate growth rates are significantly higher than salt precipitation rates and can reach on the order of  $10^6 \mu\text{m}^2/\text{min}$  for the rapid crystal growth under water-rich conditions. The mechanisms by which these two solid phases impair pore connectivity differ: salt

precipitates primarily blocking pore throats and reducing the local permeability, whereas hydrates alter fluid pathways through film formation and grain bridging. We also notice that pressure fluctuations critically affect hydrate morphology and kinetics, meaning that injection parameters need to be carefully controlled to minimize pressure fluctuations during geological CO<sub>2</sub> storage to mitigate adverse effects. Overall, salt precipitation and hydrate formation interactively influence near-wellbore reservoir performance with their relative impacts governed by reservoir thermodynamics.

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